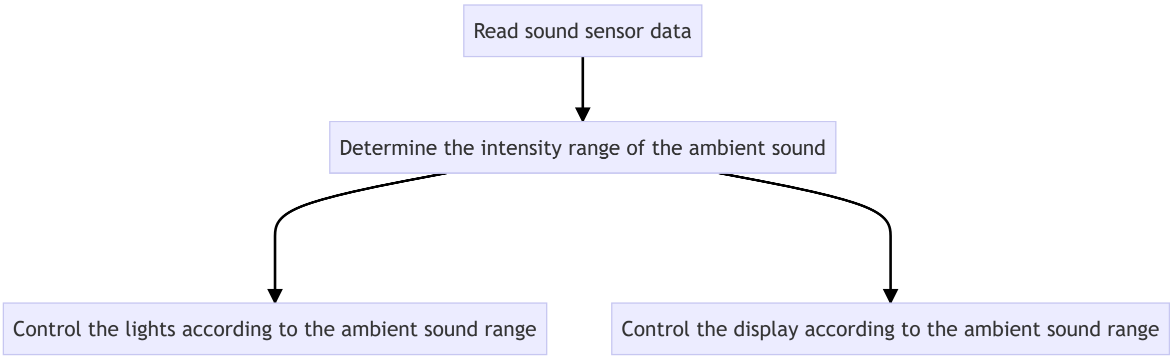
Waking You Up

1. Experiment Purpose

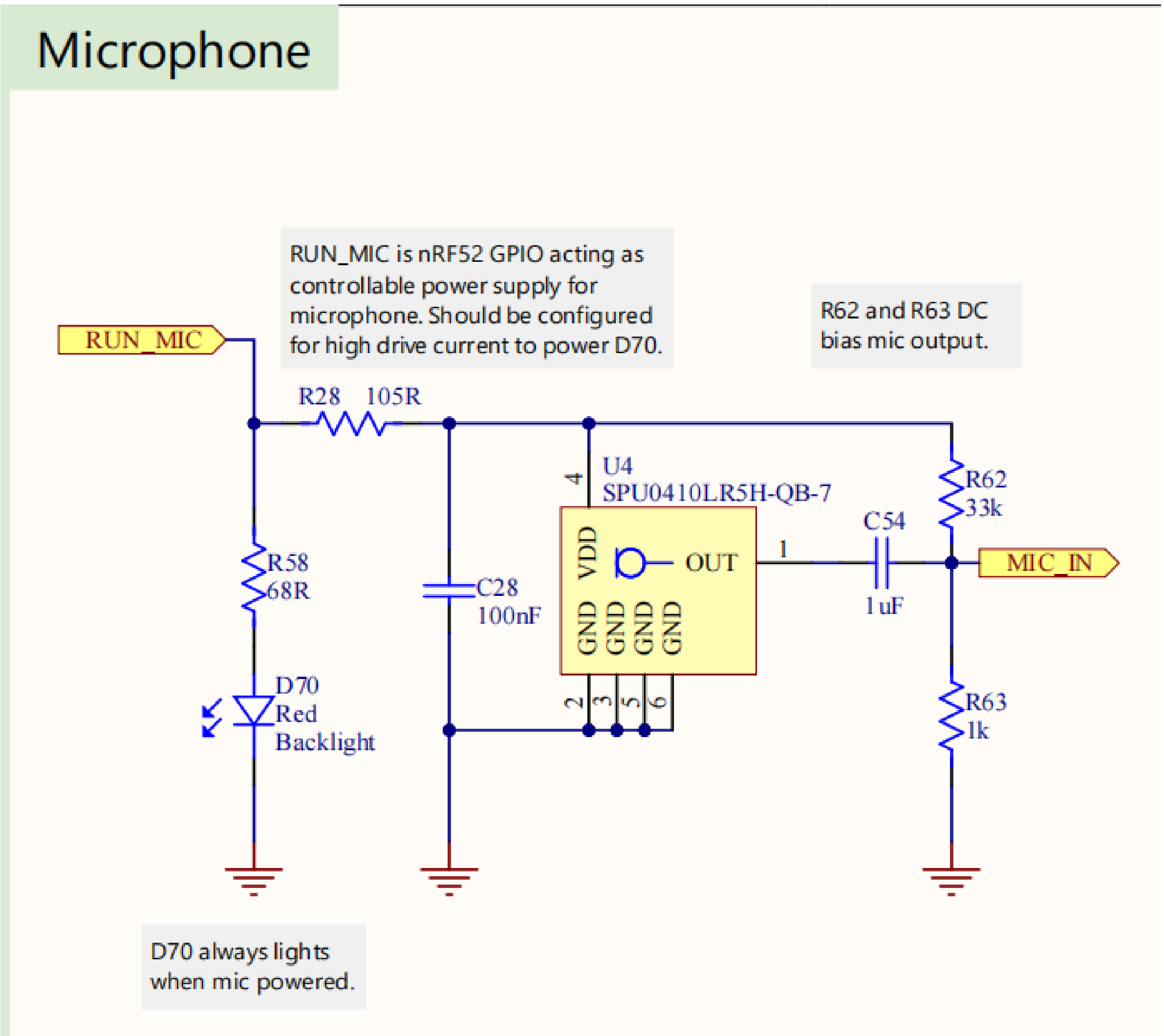
In this section, we will learn to use the microphone sensor to detect sound intensity and achieve the "waking you up" effect. The louder the sound, the brighter the lights, and the LED matrix displays bar charts of different heights to indicate the sound level.

2. Implementation Principle

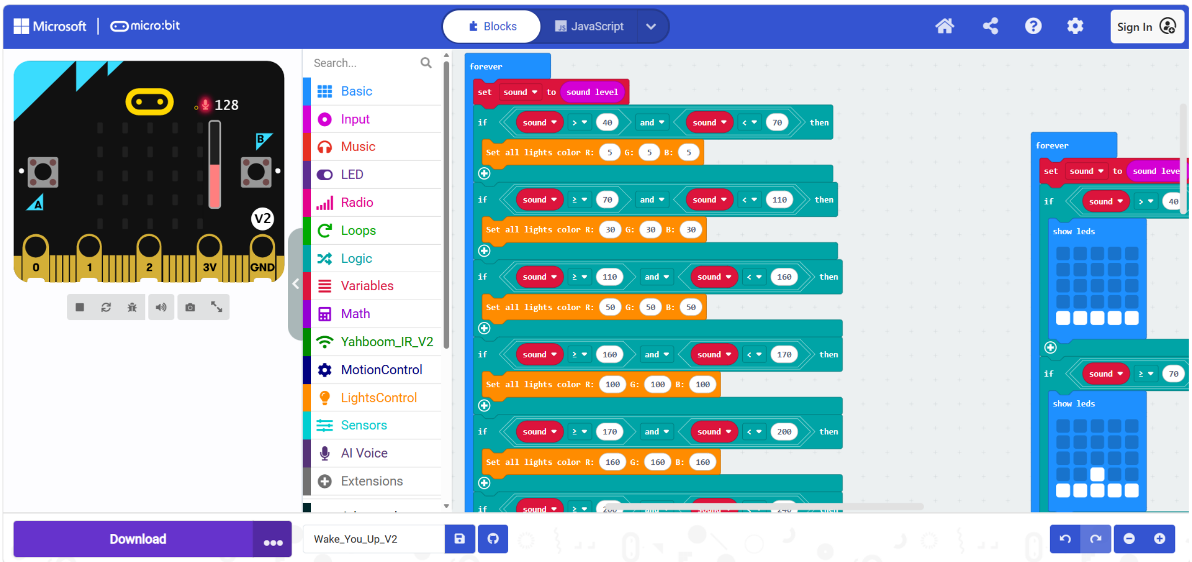


V2 version (Micro:bit V2 built-in microphone)

A microphone sensor is provided near the Logo of the Micro:bit V2. When the program drives and reads it, the red LED lights up; when it is not read, the red LED turns off and the microphone is shut down, which can reduce the power consumption of the Micro:bit at the hardware level.

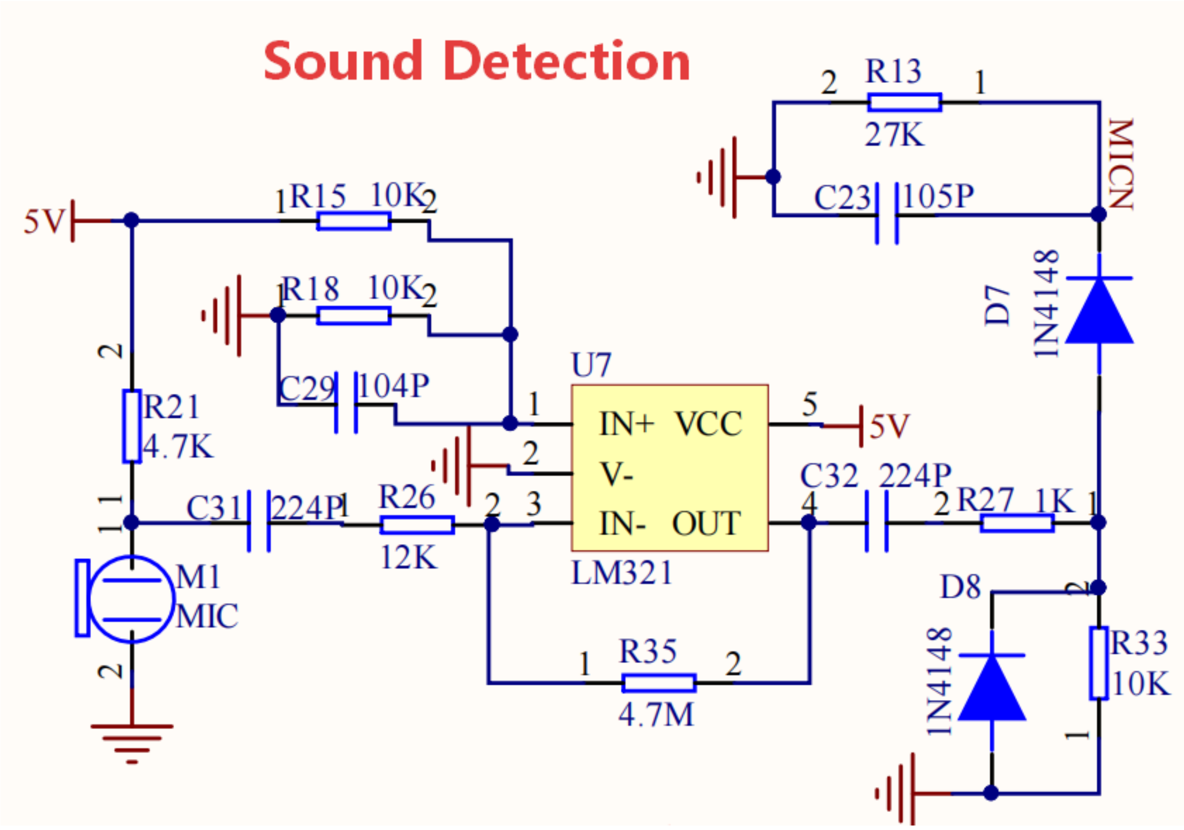


V2 version program

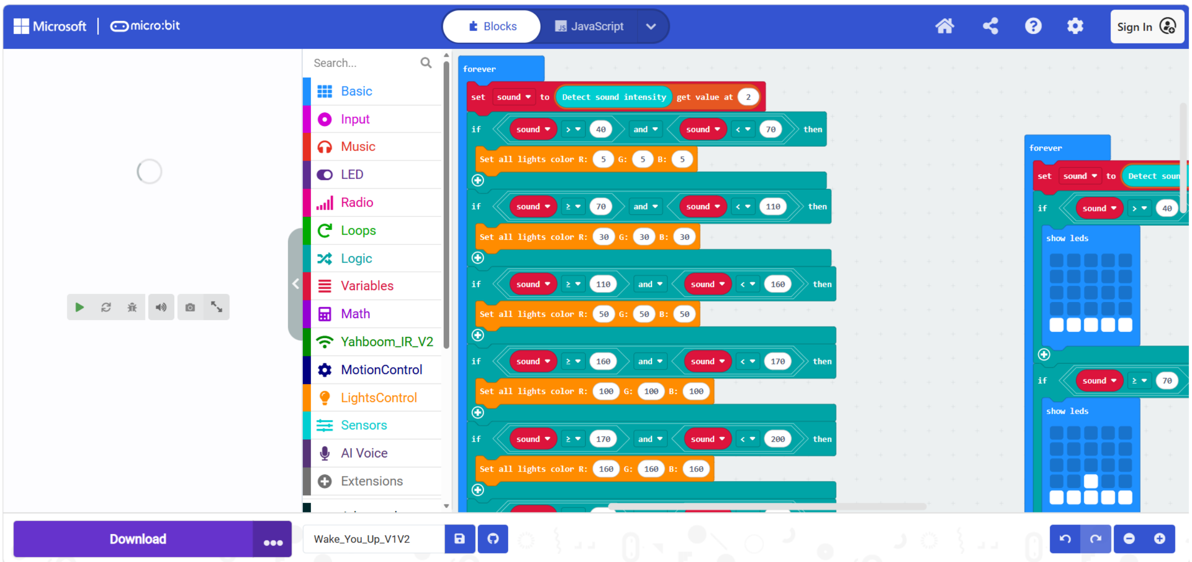


V1V2 version (expansion board MIC sensor)

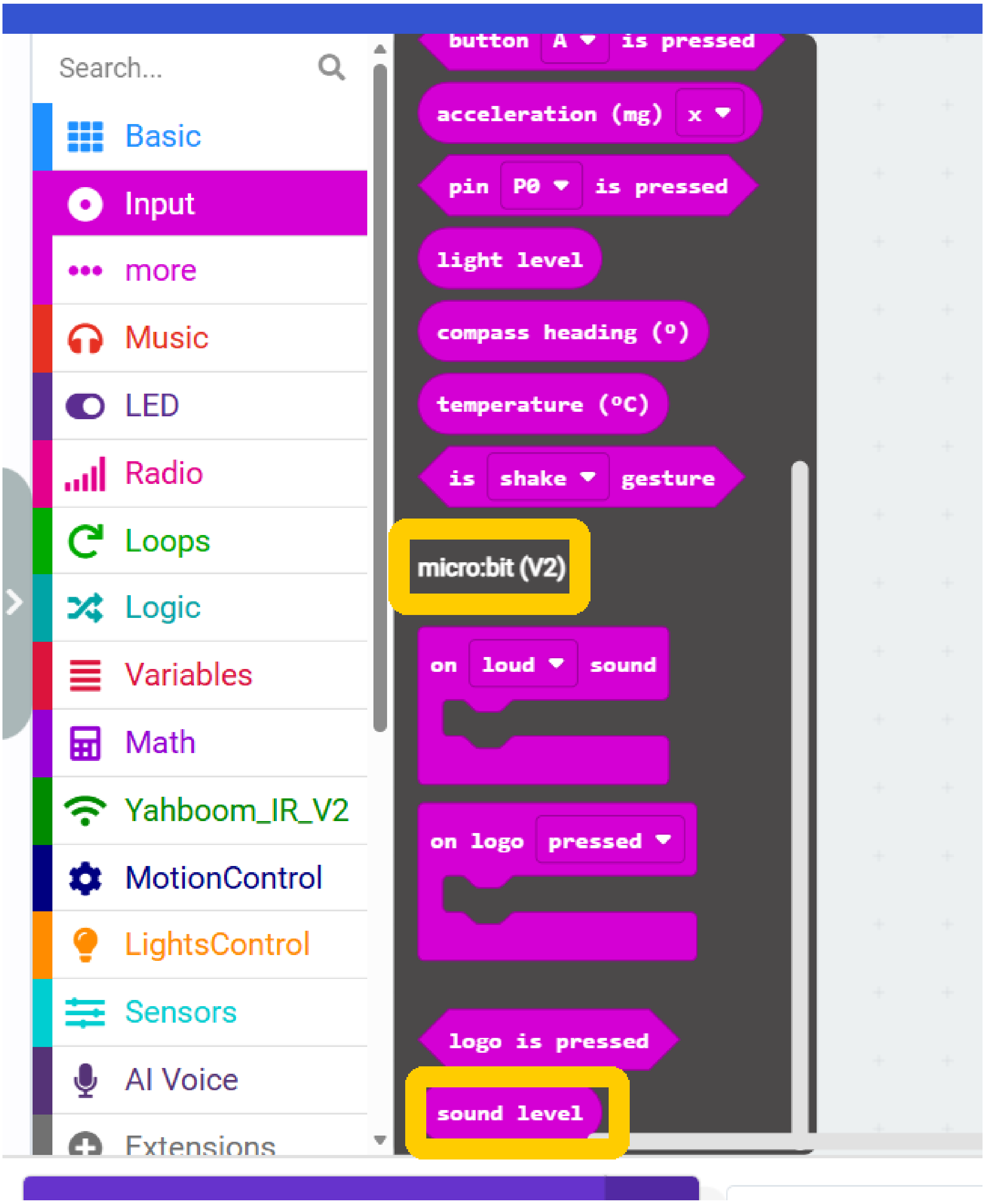
The on-board sound sensor consists of an electret microphone (commonly called a mic head) plus signal amplification and clamping protection, forming a complete sound detection circuit.



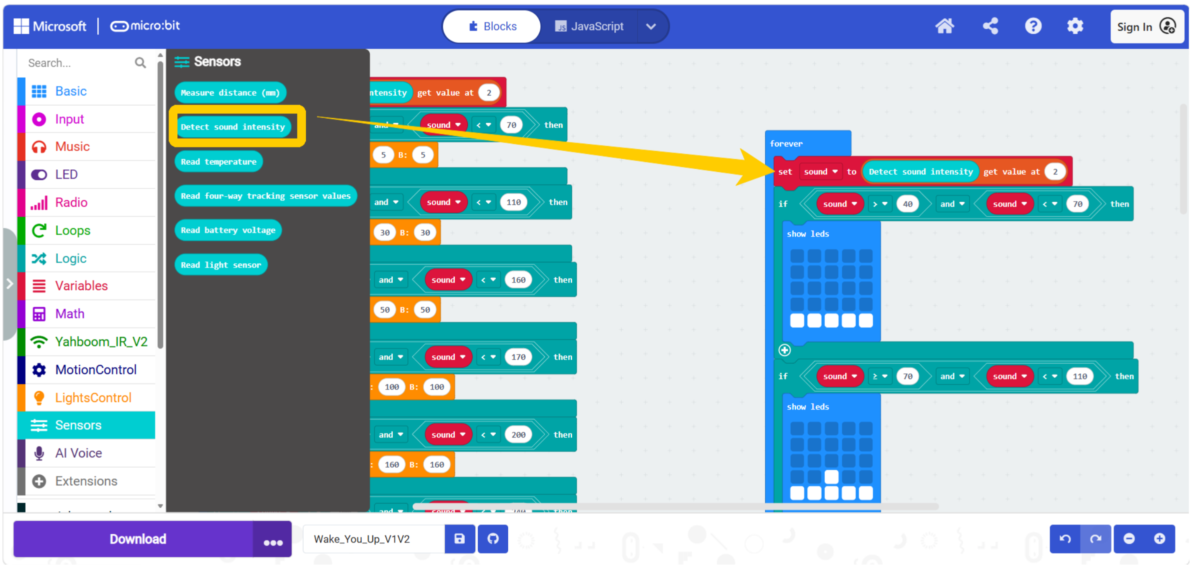
V1V2 version program



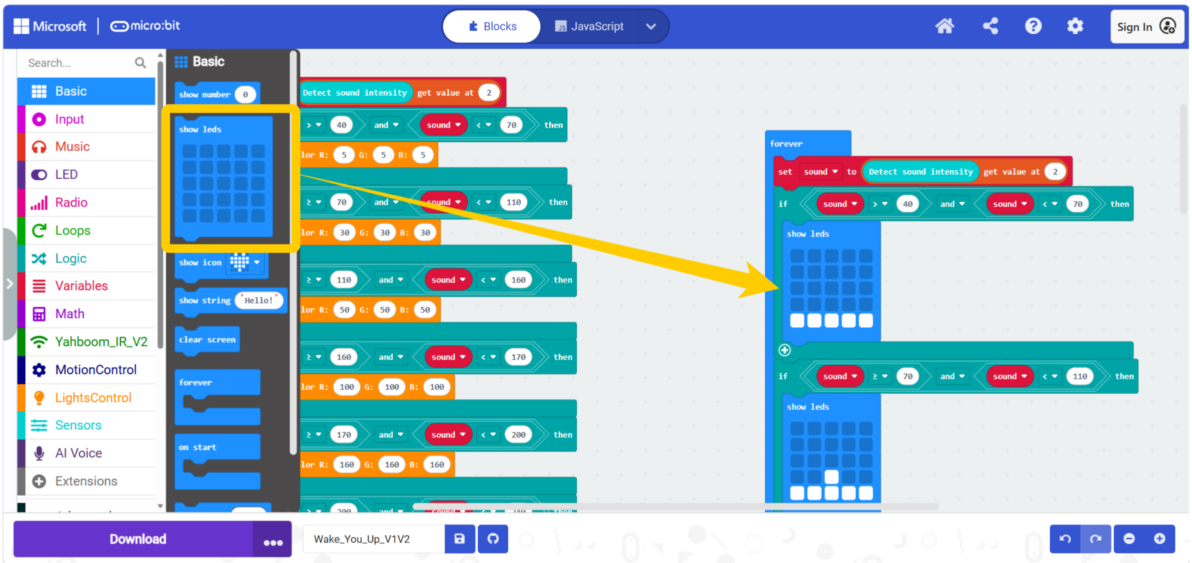
The V2 version reads the sound loudness in the `Microbit V2` subcategory of the `Input` category.



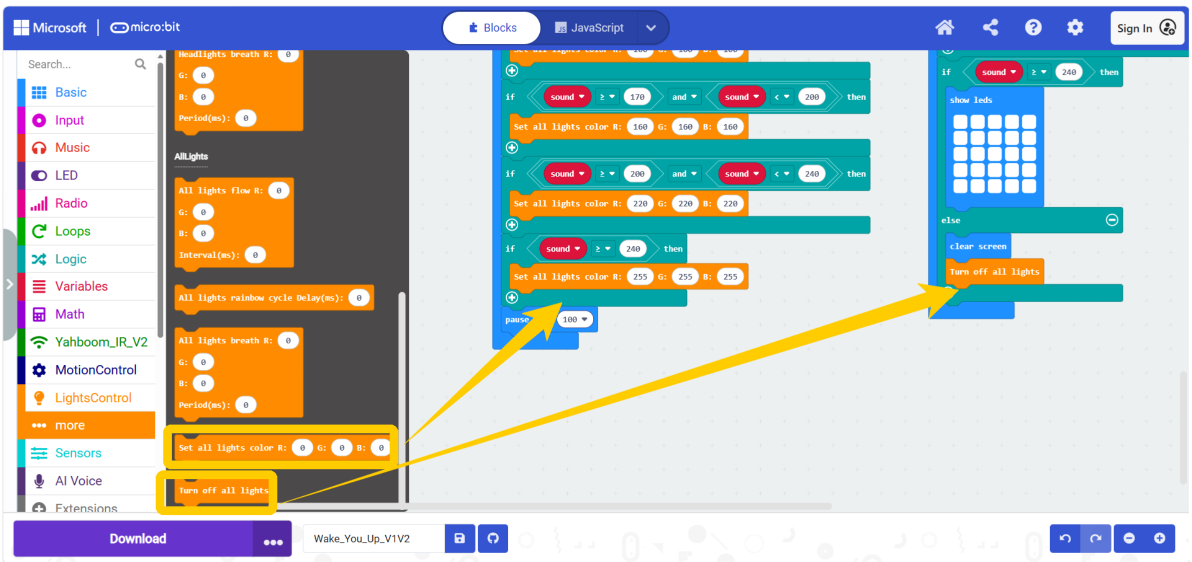
The V1V2 version reads the sound loudness in the `Sensors` category.



The block for displaying a custom pattern is located in the `Basic` category.



The light control block is located in the `more` subcategory of the `LightsControl` category.



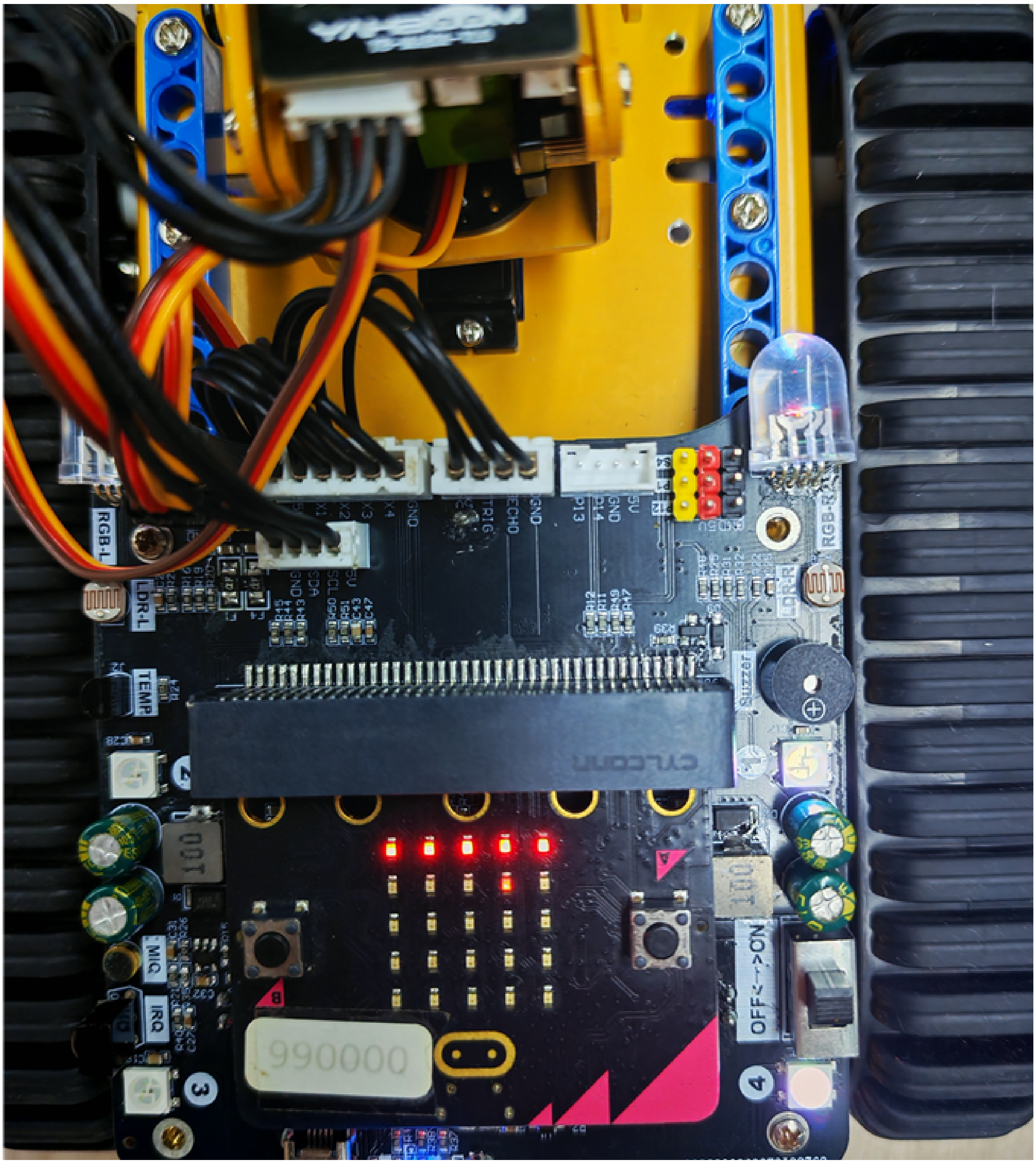
3. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.

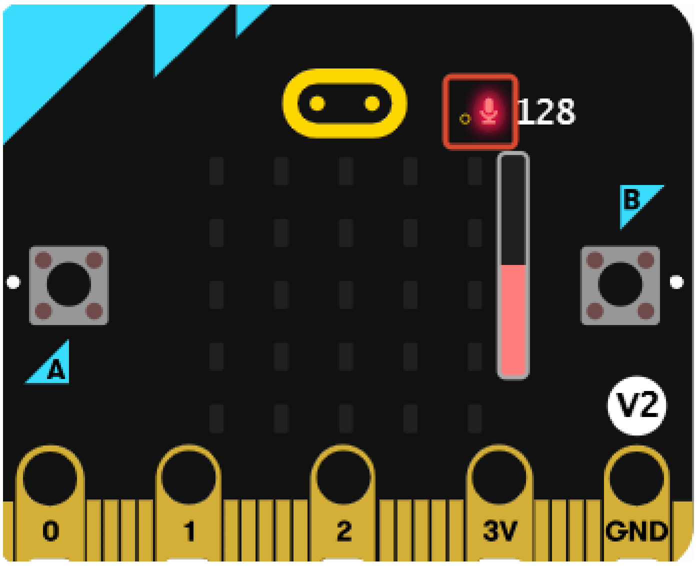
2. **Micro:bit V2 version** flashes `6.Wake_You_Up_V2.hex`, **Micro:bit V1 version** flashes `6.Wake_You_Up_V1V2.hex`. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

4. Experiment Phenomenon

After flashing and turning on the power, the car will continuously detect the ambient sound. The louder the sound, the brighter the lights, and the higher the pattern shown on the LED matrix, thus achieving the sound-controlled light effect. **Speaking near the microphone recording position makes it easier to detect the sound, and you can also adjust the sound detection threshold according to the actual situation.**



The recording position of the V2 version is as follows.



The recording position of the V1V2 version is as follows.